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RESEARCH INTERESTS

Physics-informed Dexterous Manipulation, Humanoid Learning, Reinforcement Learning

EDUCATION

Yonsei University

B.S. Candidate in Computer Science

B.S. Candidate in Nano Science and Engineering

GPA: 3.89 / 4.3

Mar. 2021 – Feb. 2027 (Expected)

(Military Service: Mar. 2023 – Sep. 2024)

West Vancouver Secondary School, Canada

International Baccalaureate Diploma

Sep. 2014 – Jun. 2020

RESEARCH

Yonsei RoPhi Lab

Advised by Prof. Yonghyeon Lee

Developing high-DoF dexterous manipulation methods with physics-informed priors to improve adaptability and generalization in unstructured environments.

Sep. 2026 – Present

KAIST HuGeLab

Summer Internship, Advised by Prof. Beomjoon Kim

Developed and deployed learning-based locomotion policies on the Unitree G1 and a custom 108 kg humanoid, with an emphasis on robustness across simulated and real-world conditions.

Jun. 2026 – Aug. 2026

Yonsei DILLAB

Advised by Prof. Jongmin Lee

Developed an offline-RL method that distills a value-aware multi-step flow policy into an efficient one-step policy using state-wise entropic optimal transport, improving multimodal policy learning across benchmark tasks.

Jan. 2026 – Jun. 2026

Yonsei RLLAB

Advised by Prof. Youngwoon Lee

Built humanoid locomotion baselines using retargeted motion priors. Analyzed how retargeting choices affect motion quality and stability, and explored reusable motion components with less task-specific reward engineering.

Jul. 2025 – Dec. 2025

Yonsei MIRLAB

Advised by Prof. Youngjae Yu

Studied multimodal learning and Isaac Sim workflows for robot learning. Contributed to an LLM scientific-verification benchmark through dataset processing, labeling, and taxonomy design.

Sep. 2024 – Mar. 2025

PUBLICATIONS

- (2) **J. Choi***, J. Shon*, J. Jeon, and J. Lee, “Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

*Equal contribution.

- (1) G. Son, J. Hong, H. Fan, H. Nam, H. Ko, S. Lim, J. Song, **J. Choi**, G. Paulo, Y. Yu, and S. Biderman, “When AI Co-Scientists Fail: SPOT—a Benchmark for Automated Verification of Scientific Research,” (2025). [arXiv:2505.11855](https://arxiv.org/abs/2505.11855).

EXPERIENCE & PROJECTS

Dexterous Manipulation Teleoperation and Learning

Advised by Prof. Yonghyeon Lee

- Developing an exoskeleton-based teleoperation system for real-time control and demonstration collection in simulation.
- Extending it toward data-driven five-finger manipulation with physics-informed priors for learned control.

Sep. 2026 – Present

Humanoid Deployment – Locomotion

Jul. 2026

Advised by Prof. Beomjoon Kim

- Trained, validated, and deployed locomotion policies on the Unitree G1 and a custom 108 kg humanoid via sim-to-sim and sim-to-real evaluation.

Franka Panda Teleoperation Data Collection

Jul. 2026

Advised by Prof. Beomjoon Kim

- Integrated a Franka Panda with a 3D mouse, Touch haptic device, and Meta Quest for teleoperated demonstration collection.

Humanoid Fixed-Base Manipulation

Mar. 2026 – Jun. 2026

Advised by Prof. Youngwoon Lee

- Investigated PPO exploration limits for humanoid fixed-base manipulation in massively parallel simulation.
- Compared PPO with SAPG and EPO and analyzed methods for improving exploration.

A1-Challenge Autonomous Driving Competition

Mar. 2026 – May 2026

Qualified in Nov. 2025; 4th Place; hosted by the Ministry of Trade and KIPI

- Built behavioral-cloning and DAGger racing policies in MORAI using targeted corner-case demonstrations.
- Explored reinforcement learning to improve robustness and driving performance.

Physical AI Robotics Hackathon – Single-Arm Manipulation

Mar. 2026

Hosted by Kernel Academy and Doosan Robotics

- Built a speech-guided pick-and-place pipeline with a Doosan E0509, RGB-D sensing, Whisper, SAM2, GPT-4o, and ROS2.
- Implemented calibrated 3D localization, task verification, and error recovery for autonomous execution.

Language-Conditioned Image Inpainting with LLMs and Diffusion Models

Jul. 2024 – Aug. 2024

OUTTA AI Bootcamp Project

- Developed a language-conditioned image editing pipeline combining LLM-based attribute extraction, semantic segmentation, and diffusion-based inpainting.

Korea Defense Intelligence Command – Translator

Mar. 2023 – Sep. 2024

Military Service

- Translated classified documents and interpreted classified meetings for KDIC and partner agencies, including the National Intelligence Service and the U.S. National Geospatial-Intelligence Agency.

VEX Robotics

Sep. 2017 – Jun. 2020

Team Lead, 1010B, WVSS Robotics

- Led a 12-member team across three seasons, overseeing strategy, mechanical and electrical design, programming, autonomous control, and competitive driving.

SELECTED TALKS & PRESENTATIONS

Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport

Jul. 2026

Research Talk, YAI (Yonsei University AI Club)

Exploration Behavior of PPO in Massively Parallel Reinforcement Learning

Jun. 2026

Poster Presentation, Yonsei UIC Science Fair

AWARDS & HONORS

Hyundai NGV Robotics Track H-Mobility Class Certificate

Mar. 2026 – Jul. 2026

Completed Hyundai's Robotics Track in robotics and robot learning and received the H-Mobility Class Certificate.

2nd Place, Physical AI Robotics Hackathon

Mar. 2026

Honor Roll, Yonsei University

Jul. 2022

1st Place, Yonsei-Nexon RC Creative Platform

Mar. 2021 – Dec. 2021

Led a five-member team to develop an automated revolving-door assistance system for people with mobility impairments; ranked 1st of 90 teams and received a \$10,000 award.

Division Finalist, VEX Robotics World Championship – Kentucky, USA

Nov. 2019

Reached the Science Division final in a 100-team division within a field of 600 qualified teams.

1st Place, VEX Robotics Canada BC Provincial Championship

Jun. 2019

Won the British Columbia Provincial Championship and qualified for the VEX Robotics World Championship.